

Vincenzo Marcianò

PhD in AI & Computer Science · Sorbonne Université / EURECOM

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📍 Vincenzo Marcianò 📍 Turin, Italy

PROFILE

I received my PhD in AI & Computer Science from Sorbonne Université and EURECOM (3IA Côte d'Azur, with King's College London) on 25 September 2026. My research asks how learned representations can tell a model when it is wrong under distribution shift: I developed generative, retrieval-augmented and distribution-level methods that assess and correct 3D medical image segmentations without ground truth, across organs, modalities and unseen datasets. I have also collaborated with Amazon, two AI start-ups and Harvard's Kempner Institute, which taught me to build systems that hold up in the real world. What I love most about research, though, is its environment: a lively mix of collaborations, where people with different backgrounds share ideas, challenge each other's assumptions and build on each other's work. It is a place driven by nothing but curiosity, and I believe this is what makes truly new ideas possible.

Research interests: domain generalisation and distribution shift · representation learning · 3D and volumetric learning · generative models (diffusion, flow matching) · retrieval-augmented and foundation models · reliable and explainable AI.

EDUCATION

PhD in Artificial Intelligence & Computer Science Sep 2023 – Sep 2026
Sorbonne Université, EURECOM (Data Science Dept., AI4Health Lab) · Biot, France

- Thesis: *Representation Learning for Medical Image Segmentation Quality*. Defended on 25 September 2026.
- Advisors: Prof. Maria A. Zuluaga (EURECOM), Prof. Sébastien Ourselin (King's College London) and Dr. Michela Antonelli (King's College London).
- Committee: Prof. Wenjia Bai (Imperial College London) and Prof. Aasa Feragen (DTU), reviewers; Prof. Bernhard Kainz (FAU Erlangen-Nürnberg) and Dr. Enzo Ferrante (Univ. of Buenos Aires), examiners.
- Doctoral funding from the 3IA Côte d'Azur Interdisciplinary Institute for Artificial Intelligence.

Visiting Researcher Sep 2023 – Sep 2026
King's College London, School of Biomedical Engineering & Imaging Sciences · London, UK

MSc in Data Science & Engineering Oct 2020 – Dec 2022
Politecnico di Torino · Turin, Italy

- Thesis: *Machine Learning and Deep Learning models to discern indoor from outdoor environments based on data recorded by a tri-axial digital magnetic sensor (published in IEEE Sensors Journal)*.
- Thesis research carried out as an exchange student at the University of Sheffield, UK (Feb – Jul 2022). Advisors: Prof. Andrea Cereatti (Politecnico di Torino), Stefano Bertuletti, Claudia Mazzà (The University of Sheffield, INDIVI).
- Involved as Lead AI Engineer in H2politO for Autonomous Driving

BSc in Electronics Engineering Oct 2017 – Jul 2020
Politecnico di Torino · Turin, Italy

- Thesis in digital filters directly coded on I/O addresses in C/Python.
- Taught as Teaching Assistant in the same electronics courses I had attended (selected for outstanding marks in the field).

RESEARCH EXPERIENCE

Doctoral Researcher

Sep 2023 – May 2026

EURECOM, AI4Health Lab (Prof. M. A. Zuluaga)  · Biot, France

- **nnQC** (IEEE TMI): a self-configuring quality-control framework in which a conditional latent diffusion model learns the “quality manifold” of an organ and scores a prediction by its distance from a generated pseudo ground truth. A Team-of-Experts module fuses image features with 3D spatial context. Evaluated on 15 datasets, 7 organs and 4 imaging techniques. It reaches a correlation of 0.89 with true Dice across 12 datasets (vs. 0.77 for the strongest prior method) with no per-dataset tuning. 
- **GRACE** (under review, TMLR): retrieval-augmented correction that moves anatomical knowledge from model weights into a multimodal knowledge base (MedGemma embeddings). A single anatomy-agnostic refiner corrects masks across 10 organs and 18 datasets, gains +0.35 Dice on weak predictions while leaving good ones untouched, and generalises zero-shot to unseen organs and datasets. It also recovers patient-specific priors across longitudinal visits without temporal supervision.
- **MID** (NeurIPS 2026): a reference-free, FID-inspired metric that compares joint image–mask distributions in a frozen medical feature space via the Wasserstein distance. It detects distribution and modality shift, ranks segmentation models as Dice does without per-sample labels, and runs in under a minute on CPU (13 datasets, 56 organ groups, 3 modalities).
- Contributed to academic collaborations by developing the coding frameworks on multimodal learning with missing modalities (HyperMM), 3D cerebrovascular annotation (VesselVerse), vascular graph extraction, and fairness-aware breast tumour segmentation.
- Supervised research projects involving **three MSc Thesis** and **two Internships**.

Research Scientist Intern

May 2022 – Sep 2022

Kroto Research Institute, University of Sheffield  · Sheffield, UK

- Improved indoor/outdoor localisation accuracy by 15% with a lightweight ML system based on time–frequency magnetometer features for embedded, wearable devices (IEEE Sensors Journal).
- Found significant gait–pathology correlations (Spearman’s $r = 0.42$) by engineering contextual time-series features that separate healthy and pathological motion. 

TEACHING AND ACADEMIC SERVICE

- **Teaching Assistant**, *Machine Learning and Intelligent Systems* (MALIS, Prof. M. A. Zuluaga), EURECOM (2023 – 2026). Holder of the lab projects for three academic years.
- **Teaching Assistant**, LED Laboratories, Politecnico di Torino (2019 – 2022). Tutorials and assessments in Applied Electronics, Biomedical Computer Architectures and Electromagnetic Fields.
- **Reviewer** for *Medical Image Analysis* (MedIA), *IEEE Transactions on Medical Imaging* (TMI) and MICCAI.
- **PhD student representative**, doctoral school, Sorbonne Université / EURECOM.

PUBLICATIONS

Own name in bold. Full list on [Google Scholar](#).

Journal Articles

- [J1] **V. Marcianò**, H. Chaptoukaev, V. Fernandez, M. J. Cardoso, S. Ourselin, M. Antonelli, M. A. Zuluaga. *Diffusion-Based Quality Control of Medical Image Segmentations across Organs*. **IEEE Transactions on Medical Imaging (TMI)**, 2026. [↗](#)
- [J2] **V. Marcianò**, A. Cereatti, S. Bertuletti, T. Bonci, L. Alcock, E. Gazit, N. Ireson, et al. *Discriminating Between Indoor and Outdoor Environments During Daily Living Activities Using Local Magnetic Field Characteristics and Machine Learning Techniques*. **IEEE Sensors Journal**, 25(1), 2024. [↗](#)

Conference Papers

- [C1] **V. Marcianò**, X. Zhang, M. Antonelli, S. Ourselin, M. A. Zuluaga. *MID: Mask-Image Distributional Divergence for Evaluating Medical Image Segmentation*. **NeurIPS 2026**.
- [C2] A. Scavone, L. Borrego, **V. Marcianò**, C. Mata, B. Giraldo, J. Munuera, M. A. Zuluaga. *Structural Congruence Matters: Biological Pretraining for Vascular Graph Extraction*. **MICCAI 2026**, Strasbourg, France. [↗](#)
- [C3] D. Falcetta, **V. Marcianò**, K. Yang, J. Cleary, L. Legris, M. D. Rizzaro, I. Pitsiorlas, et al. *VesselVerse: A Dataset and Collaborative Framework for Vessel Annotation*. **MICCAI 2025**, Daejeon, South Korea. [↗](#)

Workshop Papers

- [W1] E. Poeta, L. Vargas, D. Falcetta, **V. Marcianò**, E. Pastor, T. Cerquitelli, E. Baralis, M. A. Zuluaga. *Divergence-Aware Training with Automatic Subgroup Mitigation for Breast Tumor Segmentation*. Deep Breast Workshop, **MICCAI 2025**. **[Best Paper Award]** [↗](#)
- [W2] H. Chaptoukaev, **V. Marcianò**, F. Galati, M. A. Zuluaga. *HyperMM: Robust Multimodal Learning with Varying-sized Inputs*. MMMI Workshop, **MICCAI 2024**. **[Best Presentation Award]** [↗](#)

Under Review and Preprints

- [P1] **V. Marcianò**, X. Zhang, M. Antonelli, S. Ourselin, M. A. Zuluaga. *Retrieval-Augmented Correction for Generalist Medical Image Segmentation (GRACE)*. Under review, **Transactions on Machine Learning Research (TMLR)**.
- [P2] M. Viglino, N. V. Barrera, H. Chaptoukaev, M. A. Zuluaga, **V. Marcianò**. *Condition, Don't Impute: Missing-Aware Conditioning for Incomplete Multimodal Healthcare Data*. Preprint, 2026.

Theses

- [T1] **V. Marcianò**. *Representation Learning for Medical Image Segmentation Quality*. PhD Thesis, Sorbonne Université, 2026.
- [T2] **V. Marcianò**. *Machine Learning and Deep Learning Models to Discern Indoor from Outdoor Environments Based on Data Recorded by a Tri-axial Digital Magnetic Sensor*. MSc Thesis, Politecnico di Torino, 2022.

AWARDS AND HONOURS

2025	Best Paper Award , Deep Breast Workshop at MICCAI 2025, for <i>Divergence-Aware Training with Automatic Subgroup Mitigation for Breast Tumor Segmentation</i> .
2024	Best Presentation Award , MMMI Workshop at MICCAI 2024, for <i>HyperMM: Robust Multimodal Learning with Varying-sized Inputs</i> .
2023	Doctoral funding , 3IA Côte d’Azur Interdisciplinary Institute for Artificial Intelligence.
2022	Selected among the Nova Top 111 Students (Italian talent programme).
2019–2022	Merit-based lab-assistant scholarships , LED Laboratories, Politecnico di Torino.
2017–2022	Merit-based study scholarships , Politecnico di Torino, for the full BSc and MSc.

OPEN-SOURCE CONTRIBUTIONS

- **Overcomplete** [↗](#), Kempner Institute, Harvard (T. Fel et al.): a vision-based sparse-autoencoder toolbox for explainable AI. Contributed to the official implementation of *Relaxed Archetypal SAE* (RA-SAE) [↗](#), aligning the code with the constraints of the published method.
- **nnQC** [↗](#): lead developer. Self-configuring diffusion-based quality control for medical image segmentation, released with pretrained checkpoints and an interactive project page.
- **HyperMM** [↗](#) and **VesselVerse** [↗](#): contributor to open frameworks for multimodal learning with missing modalities and for collaborative 3D vessel annotation.

DEMOS, TALKS AND OUTREACH

- **World AI Cannes Festival (WAICF)**, Cannes, France. Live demo with **3IA Côte d’Azur** at two editions, on why representation learning matters for medical imaging and for quality control of AI segmentations.
- **Fête de la Science**, Antibes, France. Public demo of our AI research for a general audience.

INDUSTRIAL EXPERIENCE

Applied Scientist II Intern Jun 2026 – present
Amazon, Alexa AI [↗](#) · Turin, Italy

- Building benchmarks that stress-test agentic LLM routing under non-stationarity and adversarial shift. They reveal performance drops of up to 30% in state-of-the-art routers. Manuscript submitted to ICLR 2027.
- Designed a multi-armed-bandit router for shifting traffic, reducing API calls by 10% and regret by 10%.

Data & Research Scientist Jan 2026 – May 2026
LexSA (legal-tech start-up) [↗](#) · Milan, Italy

- Scraped and embedded over 1.2 million public legal documents. Built LLM-based agentic tools for fiscal profiling, improving retrieval accuracy by 50%.

AI Sound Engineer Dec 2025 – May 2026
SoundBerry (audio AI start-up) [↗](#) · Milan, Italy

- Designed transformer models for multi-stem music source separation and a

hypernetwork-based audio-to-MIDI pipeline (+10% over baseline).

Applied Scientist I Intern

Sep 2022 – Mar 2023

Amazon, Supply Chain Optimization · Barcelona, Spain

- Reduced forecasting error (WAPE) by 30% with time-domain embeddings for multivariate time-series forecasting.
- Generalised forecasting models across heterogeneous data granularities in large-scale supply-chain operations.

VOLUNTEERING AND LEADERSHIP

- **Head of Tech**, Vision APS (2025 – present). Non-profit offering scholarships, mentorship and career access to talented students facing social and geographic barriers.
- **Vice President**, Seele APS. Cultural association promoting independent festivals in Salento, Italy.
- **University Ambassador**, Nova (2021 – present).
- **Volunteer member**, ADMO, the Italian Bone Marrow Donors Association (2019 – present).

TECHNICAL SKILLS

Programming	Python, C, C++, Bash, $\LaTeX$
Deep learning	PyTorch, MONAI, TorchIO, HuggingFace, vLLM, TensorFlow, scikit-learn
Models & methods	Latent diffusion (DDIM, flow matching, DiT), VAEs, VLMs and medical foundation models (MedGemma, MedSigLIP, BiomedCLIP, DI-NOv2), LLMs, RAG and Graph-RAG, sparse autoencoders, hypernetworks, bandits
3D & robotics	3D volumetric segmentation (CT, MRI, ultrasound), ROS, NVIDIA Jetson, CARLA simulator, ITK-SNAP
Infrastructure	HPC clusters, CUDA, Docker, Git/GitHub, Linux, AWS
Certifications	AWS Machine Learning (2021); Introduction to Genomic Technologies, Johns Hopkins University (2020)
Languages	Italian (native), English (fluent), French (intermediate), Spanish (basic)

REFERENCES

Prof. Maria A. Zuluaga	Prof. Sébastien Ourselin	Dr. Michela Antonelli
PhD advisor	PhD co-director	PhD co-supervisor
EURECOM, France	King's College London, UK	King's College London, UK